

COLLEGE APPLICATION

PROJECT OUTLINE

December 11, 2025



OVERVIEW

1. Project Title and Objective

✓	Title: Echoes of Equity: Mitigating Demographic Bias in Trustworthy Intent Recognition Using Adversarial Deep Learning The objective of this project is to mitigate demographic bias and enhance classification accuracy in trustworthy intent recognition, addressing the limitations of current models that struggle with demographic generalizability. Leveraging a diverse dataset of 1,152 audio utterances from 96 speakers across Black, South Asian, and White backgrounds, this study seeks to outperform the existing 71% baseline accuracy established by traditional Random Forest models. By implementing a novel adversarial deep learning architecture, the model will be trained to disentangle universal "trustworthiness" features from demographic indicators (such as ethnicity and age), explicitly targeting the performance disparities observed in minority groups. This approach aims to create a fairer, more robust AI system that identifies trustworthy intent based on prosody rather than speaker identity.
---	---

2. Project Background

✓	Voice acoustics play a fundamental role in human interaction, influencing how we perceive personality traits such as trustworthiness. However, existing research in this field suffers from a significant "white western individualist bias" (WWIB), relying predominantly on datasets of younger, white speakers. This lack of diversity limits the generalizability of speech technologies and risks embedding demographic biases into AI systems. A recent study attempting to classify "trustworthy intent" in a demographically diverse dataset achieved only ~70% accuracy using standard machine learning models, with notable performance variations observed between White, Black, and South Asian speakers. This project addresses these critical gaps by investigating advanced deep learning architectures to improve classification accuracy and ensure equitable performance across diverse speaker demographics.
---	--

3. Project Description

✓	This project addresses a critical "White Western Individualist Bias" in voice acoustics research, where existing models for detecting trustworthiness often fail to generalize across diverse demographics. Building on a recent open-access dataset of 1,152 utterances from White, Black, and South Asian speakers, this research aims to supersede the current baseline accuracy of
---	---

	~71% while eliminating the performance disparities observed between ethnic groups. By transitioning from the paper's manual feature extraction (e.g., pitch and jitter) to a Deep Learning approach, the project seeks to create a "fairness-aware" AI that identifies universal acoustic cues of trust rather than relying on demographic-specific patterns. To demonstrate real-world utility, the final model will be deployed in a user-friendly application that provides real-time feedback on vocal trustworthiness for public speaking training. Key Features • Adversarial Debiasing: A dual-objective neural network that maximizes accuracy in detecting "trustworthy intent" while explicitly minimizing its ability to guess the speaker's age or ethnicity, ensuring the model focuses on intent rather than identity. • Raw Audio Processing: Utilizes Mel-Spectrograms generated from the raw .wav files instead of pre-calculated statistics like HNR or LTAS, allowing the AI to discover subtle prosodic features missed by traditional Random Forest models. • Demographic Parity Analysis: A specific evaluation framework designed to close the recall gap between White speakers (73%) and South Asian speakers (61%) identified in the source paper. • "TrustTrainer" Application: A lightweight, interactive desktop app that records user speech and visualizes their "trustworthiness score" in real-time, offering actionable feedback to improve communication skills. Steps to Complete • Data Acquisition & Preprocessing: Download the 1,152 normalized audio files from the Open Science Framework (OSF) and convert the signals into visual Mel-Spectrograms. • Baseline Replication: Replicate the original study's Random Forest model using the provided Python scripts to confirm the 71% accuracy benchmark. • Deep Learning Implementation: Train a Convolutional Neural Network (CNN) on the spectrogram data to establish a new "high-performance" baseline. • Bias Mitigation: Retrain the model using an "adversarial head" that penalizes the network when it successfully predicts demographic data (Ethnicity/Age), forcing it to unlearn these biases. • Application Development: Build a simple Python GUI (using Streamlit or Tkinter) that integrates the trained model, allowing users to record a sentence and receive an immediate "Trust Score" and "Fairness Check." • Final Evaluation: Compare the new model's F1-scores and AUC values against the original paper's results, specifically highlighting improvements in the Black and South Asian subgroups.
--	---

4. Project Scope

✓	This project will have two phases with overlapping tasks:
---	--

Phase 1: Learning Required Knowledge and Skills

- **Understanding Domain Science:**
 - Read the source paper thoroughly to understand the "White Western Individualist Bias (WWIB)" and the limitations of current datasets which primarily feature white, younger adults.
 - Study the acoustic features used in the paper (Fundamental Frequency/Pitch, Jitter, Shimmer, HNR, Cepstral Peak Prominence) and how they relate to perceived trustworthiness.
 - Learn about "Trustworthy Intent" vs. "Neutral Intent" and how the dataset authors collected this data using untrained speakers.
- **Mastering the Tools:**
 - **Python Basics:** Ensure comfort with core libraries (pandas for CSV handling, numpy for math).
 - **Audio Processing:** Learn to use **Librosa** or **TorchAudio** to load .wav files and convert them into Mel-Spectrograms (visual representations of audio).
 - **Machine Learning Foundations:** Learn how scikit-learn is used for Random Forest and Logistic Regression (the methods used in the paper).
 - **Deep Learning Frameworks:** Gain proficiency in **PyTorch** or **TensorFlow/Keras** to build Convolutional Neural Networks (CNNs).
- **Reproducing Baseline:**
 - Download the dataset and Python scripts from the Open Science Framework (OSF) repository mentioned in the paper.
 - Run the provided main.py script to replicate the paper's ~71% accuracy using Random Forest. This serves as the "Control" for the experiment.

Phase 2: MVP (Minimum Viable Product) Development

- **Data Pipeline Construction:**
 - Build a script to automate the loading of the 1,152 audio files and their corresponding demographic labels (Ethnicity, Age, Sex).
 - Implement data augmentation (e.g., adding slight noise or shifting pitch) to artificially increase the dataset size and improve model robustness.
- **Core Model Development (The Innovation):**
 - **Architecture Design:** Design a 2D-CNN architecture that takes Mel-Spectrograms as input to predict "Trustworthy" vs. "Neutral."
 - **Adversarial Network Setup:** Implement the "Debiasing" mechanism: add a secondary output head that tries to guess the speaker's ethnicity. Invert the gradient during training so the main model learns *not* to encode ethnicity information.

	○ Training & Tuning: Train the model to beat the 71% accuracy baseline while monitoring performance across White, Black, and South Asian subgroups. • Application & User Interface (The Product): ○ Backend Integration: Save the trained PyTorch/Keras model as a portable file (e.g., .h5 or .pt). ○ Frontend Creation: Develop a simple GUI using Streamlit (Python). ▪ <i>Feature A:</i> A "Record" button that captures 3 seconds of audio. ▪ <i>Feature B:</i> A visual display showing a "Trust Score" (0-100%). ▪ <i>Feature C:</i> A "Feedback" text box suggesting vocal adjustments (e.g., "Try steadying your pitch"). • Final Validation & Metrics: ○ Generate a Confusion Matrix and ROC Curve to compare against the paper's Table 9 and Table 10. ○ Explicitly measure the "Fairness Gap" (difference in accuracy between demographic groups) to prove the project achieved its goal of equity.
--	--

5. Project Milestones

✓	The following are the milestones of the project. Phase 1: Learning & Baseline Verification • Milestone 1: Literature & Dataset Audit ○ Complete a deep reading of the paper, specifically analyzing the "White Western Individualist Bias (WWIB)" and the performance gaps found in Black and South Asian speakers. ○ Download the full dataset of 1,152 audio utterances from the Open Science Framework (OSF) repository. • Milestone 2: Environment Configuration ○ Set up a local Python development environment with the necessary dependencies (e.g., scikit-learn, pandas) as listed in the repository's requirements.txt. ○ Install advanced audio-processing libraries (like Librosa) and Deep Learning frameworks (like PyTorch or TensorFlow). • Milestone 3: Baseline Replication (The Control Group) ○ Run the provided main.py script to successfully train the Random Forest model and reproduce the paper's reported ~71% accuracy.
---	--

	○ Document the specific "Recall" and "Precision" scores for each ethnicity to serve as the benchmark to beat. Phase 2: Innovation & MVP Development • Milestone 4: Data Engineering Pipeline ○ Create a script to convert the raw .wav files into Mel-Spectrograms (visual heatmaps of sound), moving away from the paper's manual feature extraction. ○ Implement data augmentation (adding noise/pitch shifting) to robustly train the model on the relatively small dataset of 96 speakers. • Milestone 5: Deep Learning Model Training ○ Design and train a Convolutional Neural Network (CNN) that achieves a higher overall accuracy than the paper's 71% baseline. ○ Verify that the model works on raw audio inputs rather than just the CSV of acoustic features. • Milestone 6: Bias Mitigation (The Core Innovation) ○ Implement "Adversarial Debiasing" by adding a second output head to the model that attempts to predict the speaker's demographics (Age/Ethnicity). ○ Retrain the model to minimize demographic prediction success while maximizing trust detection, effectively closing the performance gap between White and South Asian speakers. • Milestone 7: "TrustTrainer" App Development ○ Build a functional GUI (using Streamlit) that allows a user to record their voice via microphone. ○ Display a real-time "Trustworthiness Score" and visual feedback based on the model's inference. • Milestone 8: Final Analysis & Presentation ○ Generate new Confusion Matrices and ROC Curves to directly compare against the paper's Table 9 and Table 10. ○ Finalize the ISEF display board, highlighting the reduction in demographic bias as the primary scientific contribution.
--	--

6. Deliverables

✓	Software & Technical Artifacts • Source Code Repository: A complete GitHub repository containing Python scripts for data preprocessing, baseline replication (Random Forest), and the novel Adversarial Deep Learning framework.
---	---

	• Trained "Fairness-Aware" Model: The final optimized Deep Learning model file (e.g., .pt or .h5) capable of classifying trustworthy intent with higher accuracy than the paper's 71% baseline. • "TrustTrainer" Application: A functional Minimum Viable Product (MVP) using a Streamlit GUI that records user audio, visualizes real-time "Trust Scores," and offers feedback. • Spectrogram Dataset: A processed library of Mel-Spectrogram images generated from the original 1,152 .wav files, serving as the input data for the CNN. Scientific Analysis & Presentation • Comparative Performance Report: A statistical analysis contrasting the new model's F1-scores against the paper's original Random Forest results, specifically highlighting the reduced performance gap between White and South Asian speakers. • Visual Evidence Suite: A set of generated Confusion Matrices and ROC Curves to directly compare against the original study's Figures and Tables. • ISEF Research Paper: A formal 20-page scientific paper detailing the methodology (Adversarial Debiasing), results, and conclusion. • Project Display Board: A visual presentation summary including the problem statement, "Fairness" hypothesis, architectural diagrams of the neural network, and final real-world application screenshots.
--	---

PROJECT TIMELINES

We approve the project as described above and authorize the student to proceed.

Task Description	From Date	To Date
Python Basics	Dec 15, 2025	Jan 18, 2026
Data Handling (Pandas)	Jan 19, 2026	Feb 15, 2026
Math for AI (NumPy)	Feb 16, 2026	Mar 15, 2026
Intro to ML (Scikit-Learn)	Mar 16, 2026	Apr 19, 2026
Replication – Random Forest	Apr 20, 2026	May 10, 2026
Replication – Confusion Matrix	May 11, 2026	Jun 14, 2026
Audio Processing	Jun 15, 2026	Jul 12, 2026
Intro to Neural Networks	Jul 13, 2026	Aug 16, 2026
Model Innovation – Train CNN	Aug 17, 2026	Sep 13, 2026
App Building (Streamlit)	Sep 14, 2026	Oct 11, 2026

Integration – Connect DP model to the app	Oct 12, 2026	Nov 08, 2026
Data Visualization	Nov 09, 2026	Dec 06, 2026
Final Polish – Prepare for Regional Fair	Dec 07, 2026	Dec 15, 2026

PROJECT COST

Title	Cost	Time
Provide guidance in learning the required materials.	\$5000	12/15/2025
Offer mentorship and support throughout the project development.	\$4500	6/15/2025
Total	\$9,500	

AGREEMENT BETWEEN INSTRUCTOR AND STUDENT

This Agreement ("Agreement") is entered into between Duo Chen, and Owen Wu on December 15, 2025.

Responsibilities of the Student:

- The student agrees to work hard to learn the course materials, including watching videos, reading textbooks, completing assignments and working on the project.
- The student agrees to complete all quizzes and homework assignments on time.
- The student agrees to develop AI model, create the app, test and improve.
- The student agrees to ask the instructor any questions regarding the course materials or assignments.
- The student agrees to take full responsibility if they fail to meet the requirements and are unable to complete the project.

Responsibilities of the Parent:

- Remind the child to complete learning materials as per the schedule.
- Ensure the child submits homework before the deadline.
- If the child falls behind, collaborate with him and the instructor to get back on track.

Responsibilities of the Instructor:

- The instructor agrees to answer the student's questions in a timely manner.
- The instructor agrees to mentor the student to finish the project.
- The instructor will not write the code for the student in the project development.

The instructor cannot guarantee that the student will win the competition. Success depends on the student's commitment to learning the necessary skills and actively applying them to the project.

By signing below, the parties acknowledge that they have read and understand this Agreement and agree to its terms and conditions.

Duo Chen

Owen Wu

Duo Chen

Date: 12/15/2025

Date: 12/15/2025